Specifying and Analyzing Transactional Consistency Models with Predicates

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A semantic and graph-theoretic foundation for predicate operations

Item Operations vs. Predicate Operations

Read(x, v)
Read$(2, 60)$

Write(x, v)
Write$(3, 40)$

point access
one value

Employee

EID	Salary
0	80
1	100
2	60
3	40

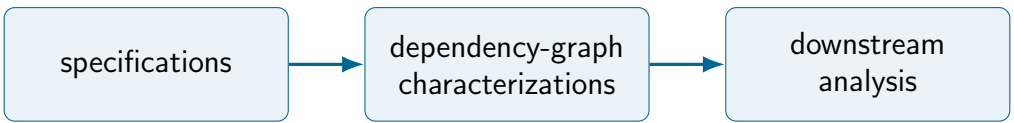
PredRead(P, M)
P : $Salary < 80$

condition-based access
a set of values

One query can observe both matching and non-matching employees.

Review

What we know about transactional consistency models without predicates



A mature correspondence for named items and item dependencies.

Review: Intuitive or Operational Specs

The Notions of Consistency and Predicate Locks in a Database System

K.P. Eswaran, J.N. Gray,
R.A. Lorie, and I.L. Traiger
IBM Research Laboratory
San Jose, California

SER

serial execution

A Critique of ANSI SQL Isolation Levels

Hal Berenson — Microsoft Corp. — haroldb@microsoft.com
Phil Bernstein — Microsoft Corp. — philbe@microsoft.com
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4.2 **Snapshot Isolation**

A transaction executing with *Snapshot Isolation* always reads data from a snapshot of the (committed) data as of the time the transaction started, called its *Start-Timestamp*. This time may be any time before the transaction's first Read. A transaction running in Snapshot Isolation is never blocked attempting a read as long as the snapshot data from its Start-Timestamp can be maintained. The transaction's writes (updates, inserts, and deletes) will also be reflected in this snapshot, to be read again if the transaction accesses (i.e., reads or updates) the data a second time. Updates by other transactions active after the transaction Start-Timestamp are invisible to the transaction.

Snapshot Isolation is a type of multiversion concurrency control. It extends the Multiversion Mixed Method described in [BHG], which allowed snapshot reads by read-only transactions.

When the transaction T1 is ready to commit, it gets a *Commit-Timestamp*, which is larger than any existing Start-Timestamp or Commit-Timestamp. The transaction successfully commits only if no other transaction T2 with a Commit-Timestamp in T1's interval [*Start-Timestamp*, *Commit-Timestamp*] wrote data that T1 also wrote. Otherwise, T1 will abort. This feature, called *First-committer-wins* prevents lost updates (phenomenon P4). When T1 commits, its changes become visible to all transactions whose Start-Timestamps are larger than T1's Commit-Timestamp.

SI snapshot + write-conflict control

Eswaran et al., 1976; Berenson et al., 1995

Review: Axiomatic Specs

A shared execution vocabulary: (VIS, AR)


A Framework for Transactional Consistency Models with Atomic Visibility

Andrea Cerone, Giovanni Bernardi, and Alexey Gotsman

IMDEA Software Institute, Madrid, Spain

atomic visibility


Seeing is Believing: A Client-Centric Specification of Database Isolation

Natacha Crooks, The University of Texas at Austin and Cornell University
Youer Pu, Cornell University
Lorenzo Alvisi, The University of Texas at Austin and Cornell University
Allen Clement, Google, Inc.

client-centric views


On the Complexity of Checking Transactional Consistency

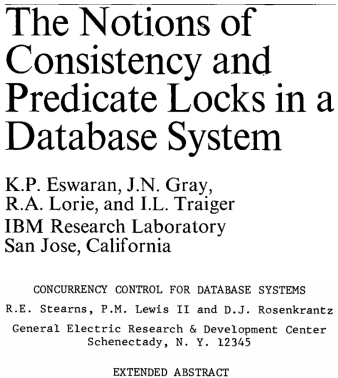
RANADEEP BISWAS, Universite de Paris, IRIF, CNRS, France
CONSTANTIN ENEA, Universite de Paris, IRIF, CNRS, France

checking complexity

Visibility determines observable item versions; arbitration orders writes.

Cerone, Bernardi, and Gotsman, 2015; Crooks et al., 2017

Review: Extract Dependency Graphs from Histories



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CONCURRENCY CONTROL FOR DATABASE SYSTEMS

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General Electric Research & Development Center
Schenectady, N. Y. 12345

EXTENDED ABSTRACT

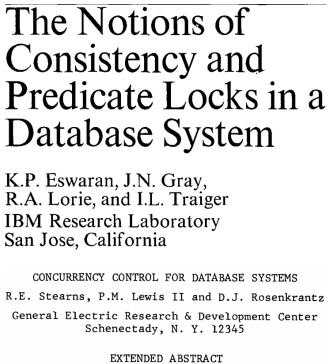
History

$\Longrightarrow$ item dependency graph $\Longrightarrow$

WR, WW, RW edges encode observed versions and overwrites.

Eswaran et al., 1976; Bernstein, Hadzilacos, and Goodman, 1987

Review: SER Characterization Theorem



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EXTENDED ABSTRACT

Theorem

An item-operation history is serializable iff its dependency graph is internally consistent and acyclic.

Bernstein, Hadzilacos, and Goodman, 1987; Adya, 1999

Review: SI Characterization Theorem

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Theorem

An item-operation history satisfies SI iff its dependency graph is internally consistent and every cycle has two adjacent RW *edges.*

This exact correspondence is the baseline that predicates must recover.

Cerone and Gotsman, 2018

Review: Downstream Tasks

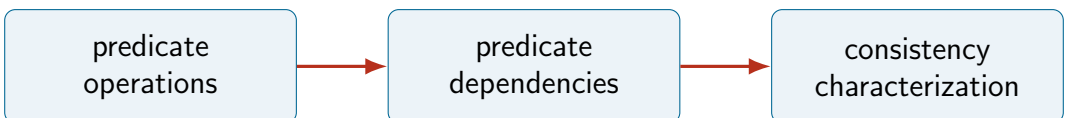
Task	Representative work
Black-box checking	Elle, COBRA, PolySI, VeriStrong
CC protocol verification	execution-based verification
Robustness	robustness against weaker consistency

All depend on a faithful link between observed histories and consistency.

Kingsbury and Alvaro, 2020; Tan et al., 2020; Huang et al., 2023; Cai et al., 2025; Bernardi and Gotsman, 2016

Review

What we know about transactional consistency models
with predicates



The question now includes which items a query did not return.

Review: Intuitive or Operational Specs

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SI snapshot + write-conflict control

Predicate semantics is not supplied by these item-level specifications.

Eswaran et al., 1976; Berenson et al., 1995

Review: Axiomatic Specs

On the Complexity of Checking Mixed Isolation Levels for SQL Transactions

Ahmed Bouajjani[1], Constantin Enea[2], and Enrique Román-Calvo[1]

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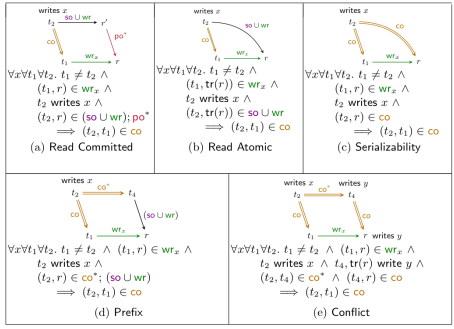
CAV Artifact Evaluation ★ Available

checking-oriented consistency conditions



axiomatic specification grammar

The challenge: make the specification explain matching and omitted items.

Bouajjani, Enea, and Román-Calvo, 2025

Review: Dependency Graphs

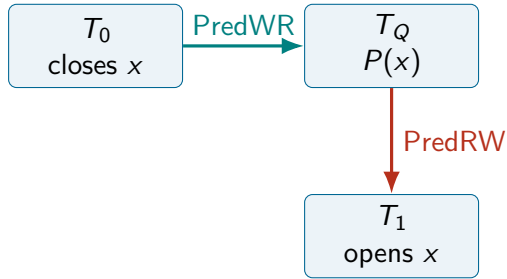
Weak Consistency: A Generalized Theory and Optimistic Implementations for Distributed Transactions

by

Atul Adya

Submitted to the Department of Electrical Engineering and Computer Science
in partial fulfillment of the requirements for the degree of

Doctor of Philosophy



A predicate read depends on a witness and on later result-changing writers.

Adya, 1999

Review: Dependency Graphs


Generalized Isolation Level Definitions

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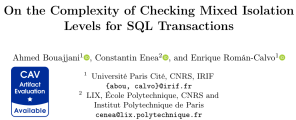
Boston, MA 02125-3393
poneil@cs.umb.edu

generalized isolation


Vbox: Efficient Black-Box Serializability Verification

Anonymous Author(s)

earliest result change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CAV Artifact Evaluation Available

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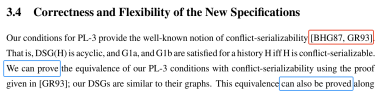
all later writers

$$\text{PredRW}^- \subseteq \text{PredRW} \subseteq \text{PredRW}^+$$

Predicate-read and predicate-anti-dependency definitions differ.

"Generalized Isolation Level Definitions" 2000; Sun and Zou, 2025; Bouajjani, Enea, and Román-Calvo, 2025

Review: SER Characterization



3.4 Correctness and Flexibility of the New Specifications

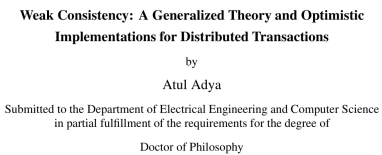
Our conditions for PL-3 provide the well-known notion of conflict-serializability [BHG87, GR93]. That is, DSG(H) is acyclic, and G1a, and G1b are satisfied for a history H iff H is conflict-serializable. We can prove the equivalence of our PL-3 conditions with conflict-serializability using the proof given in [GR93]; our DSGs are similar to their graphs. This equivalence can also be proved along the lines of the proof given in [BHG87] for view-serializability.

Item-level serializability has a well-established graph proof.

Predicate extensions use different edges, but lack a single rigorous semantic proof bridge.

Bernstein, Hadzilacos, and Goodman, 1987; Gray and Reuter, 1993; Adya, 1999; "Generalized Isolation Level Definitions" 2000; Fekete et al., 2005; Clark, Donaldson, et al., 2024; Clark, Rigger, et al., 2026; Sun and Zou, 2025

Review: SI Characterization



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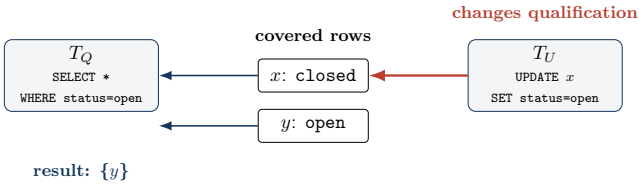
Item-level SI requires two adjacent anti-dependencies in every cycle.

Which predicate edges preserve this exact characterization?

Adya, 1999; Cerone and Gotsman, 2018

The Problem



Key observation: $x \notin$ result
still records an observation of x.

Item read: value observed

Predicate read: returned and excluded items

A phantom changes the result through absence.

Can one predicate-aware foundation support both specifications and graphs?

Berenson et al., 1995; Adya, 1999